
A Dual-branch VAE-LSTM with Multi-head Attention for Real-time Stress Detection

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Abstract

Detecting physiological stress from wearable sensors is challenging because it alters signals at multiple time scales. Existing unsupervised methods, including MA-VAE and VAE-LSTM, often struggle to provide the discriminative feature representation and long-term contextual integration. To address these limitations, we propose a dual-branch VAE-LSTM architecture utilizing a Multi-head Attention mechanism to dynamically fuse local and global features. Evaluated on the WESAD dataset, our proposed model achieves an F1 score of 0.941, outperforming existing baseline methods. This result validates that the dynamic fusion of multi-scale features is critical for resolving the challenges in stress detection.

1. Introduction

Anomaly detection identifies patterns that deviate from normal behavior (Yang et al., 2023; Li & Jung, 2023). Physiological stress can be viewed as such an anomaly because it disrupts normal function and produces measurable autonomic responses (increased HR, decreased HRV, elevated EDA) that wearable sensors capture non-invasively (Gedam & Paul, 2021; Albaladejo-González et al., 2023). Since self-reports are unreliable for continuous tracking, detecting stress directly from multi-modal physiological signals has become essential, as each modality captures complementary aspects of stress and reduces false positives (Taskasaplidis et al., 2024; Yang et al., 2023).

Although supervised learning is common, large-scale annotation is expensive and inconsistent, leading to severe label scarcity (Liu et al., 2023). High inter-subject variability

further limits generalization, and cross-dataset transfer remains unreliable (Albaladejo-González et al., 2023). These limitations motivate unsupervised approaches that detect stress as a distributional deviation without requiring labels.

This study focuses on robust unsupervised detection using the WESAD dataset, but preliminary experiments reveal fundamental limitations in established baseline architectures. While the MA-VAE model (Correia et al., 2023) benefits from its Multi-head Attention mechanism in weighing feature importance, its limited contextual scope struggles to distinguish anomalies from noise. In contrast, the prediction-based VAE-LSTM model (Lin et al., 2020) is effective in modeling temporal dynamics, yet is limited by its short-term temporal focus, leading to suboptimal coverage of anomalies that manifest across longer timescales.

This failure to capture necessary long-term contextual patterns motivates the development of a novel anomaly detection framework capable of integrating the superior predictive power with multi-scale contextual feature fusion. The primary objective of this work is to introduce a dual-branch VAE-LSTM architecture utilizing a Multi-head Attention mechanism to dynamically fuse local and global features, thereby achieving robust and high F1 score unsupervised stress detection. We hypothesize that this dynamic fusion is critical for overcoming the inherent trade-off, and we aim to validate this by demonstrating that our proposed model significantly outperforms existing baseline models.

2. Backgrounds

2.1. Variational Autoencoder (VAE)

The Variational Autoencoder (VAE) (Kingma & Welling, 2013) provides the necessary probabilistic framework for learning the underlying distribution of the ‘normal’ (non-stressed) time series data $\mathbf{X} \in \mathbb{R}^{T \times D_x}$. The VAE objective is to find optimal parameters ϕ (encoder) and θ (decoder) by maximizing the Evidence Lower Bound ($\mathcal{L}_{\text{ELBO}}$), which consists of a reconstruction term and a Kullback-Leibler (KL) divergence term.

$$\mathcal{L}_{\text{ELBO}}(\phi, \theta; \mathbf{x}) = \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})}[\log p_{\theta}(\mathbf{x}|\mathbf{z})] - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})||p(\mathbf{z})). \quad (1)$$

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Here, the encoder q_ϕ is assumed to approximate the true posterior $p(\mathbf{z}|\mathbf{x})$ as a Gaussian distribution.

$$q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_\phi(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}_\phi^2(\mathbf{x}))). \quad (2)$$

In anomaly detection, a large reconstruction error signifies a low probability under the learned normal distribution, thus classifying the input as an anomaly.

The VAE provides a stable foundation for generative modeling but often struggles to learn disentangled latent factors on complex datasets. To address this limitation, the β -VAE model was proposed to disentangle independent factors of variation in data (Higgins et al., 2017). β -VAE augments the VAE objective function by introducing a hyperparameter, β , which constrains the capacity of the latent information bottleneck. The objective function of β -VAE is defined as

$$\mathcal{L}(\phi, \theta; \mathbf{x}) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})] - \beta D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p(\mathbf{z})). \quad (3)$$

2.2. VAE-LSTM

The VAE-LSTM hybrid model is an unsupervised approach designed to capture both local features and long-term trends in time series data (Lin et al., 2020). Its training proceeds in two sequential stages: a VAE module first learns a compact latent representation of local normal patterns by maximizing $\mathcal{L}_{\text{ELBO}}$, and subsequently, an LSTM module is optimized to predict the sequence of these latent embeddings. During inference, the VAE encoder and decoder reconstructs future windows based on the LSTM's latent predictions. The final anomaly score is calculated as the sum of squared prediction errors between the actual future windows and their reconstructed counterparts. A score exceeding a predetermined threshold τ signals an anomaly.

2.3. MA-VAE

The MA-VAE architecture, an unsupervised approach leveraging a Multi-head Attention (MA) mechanism, achieves high accuracy and fast, parallelizable inference for anomaly detection (Correia et al., 2023). Its encoder q_ϕ maps the input window $\mathbf{X}$ to the latent representation $\mathbf{Z}$. The core MA layer uses $\mathbf{X}$ to derive the query $\mathbf{Q}$ and key $\mathbf{K}$ matrices, while the latent $\mathbf{Z}$ is used for the value $\mathbf{V}$, combining this information into a context matrix $\mathbf{C}$. The decoder p_θ then utilizes $\mathbf{C}$ to reconstruct the input. During inference, the anomaly score is calculated as the sum of the squared errors between the original input and its reconstruction. A score exceeding a predetermined threshold τ signals an anomaly.

3. Proposed Model

The architecture of our proposed model, illustrated in Figure 1, integrates multi-scale feature extraction with a robust

temporal prediction mechanism to detect anomalies as significant deviations between predicted and actual time series segments. The model processes the raw time series data $\mathbf{X}$ using a sliding window $\mathbf{W}_t$ and consists of four main components: a dual-branch encoder, a Multi-head Attention mechanism for feature fusion, a latent prediction with LSTM, and a decoder.

3.1. Dual-Branch Encoder

The Encoder is the initial stage of the pipeline, designed to extract distinct feature representations from the input window $\mathbf{W}_t$ across different temporal scales. This dual-branch design ensures that both broad contextual information and subtle, rapid-changing patterns are captured before fusion. The input window $\mathbf{W}_t$ is processed concurrently by two parallel modules: the Global Module and the Local Module.

3.1.1. GLOBAL MODULE

The Global Module is responsible for extracting long-term contextual features. By processing the entire window $\mathbf{W}_t \in \mathbb{R}^{L \times D_x}$ at once, it captures the overarching trends and general statistical characteristics of the input segment. This module typically employs a stack of 1D convolutional (Conv1D) layers followed by max pooling (MaxPool1D) to progressively reduce dimensionality and extract scale-invariant features. The output is a single, compact feature vector, the global embedding e_{glob}^t , which summarizes the overall content of the window $\mathbf{W}_t$.

3.1.2. LOCAL MODULE

The Local Module is dedicated to capturing short-term, fine-grained temporal embeddings. To prepare the input for this module, the input window $\mathbf{W}_t$ is first segmented into k smaller parts, w_i^t , such that $\mathbf{W}_t = [w_1^t, \dots, w_k^t]$, where each segment w_i^t has a length l and $L = k \times l$. It then takes this sequence of segmented windows $\mathbf{W}_t \in \mathbb{R}^{k \times l \times D_x}$ as input. Each segment w_i^t is processed to learn high-resolution patterns specific to that small interval, enabling the detection of subtle, abrupt changes that might be smoothed out by the Global Module. The output is a sequence of local embeddings $[e_1^t, e_2^t, \dots, e_k^t]$, where e_i^t corresponds to the detailed representation of the segment w_i^t .

3.2. Multi-head Attention for Local-Global Feature Fusion

The subsequent stage consolidates information via a Multi-head Attention (MA) mechanism for local-global feature fusion. This allows the model to dynamically weigh the overall context when processing local segments. We configure the local embeddings $[e_1^t, \dots, e_k^t]$ as the query $\mathbf{Q}$ and the global embedding e_{glob}^t as both the key $\mathbf{K}$ and value

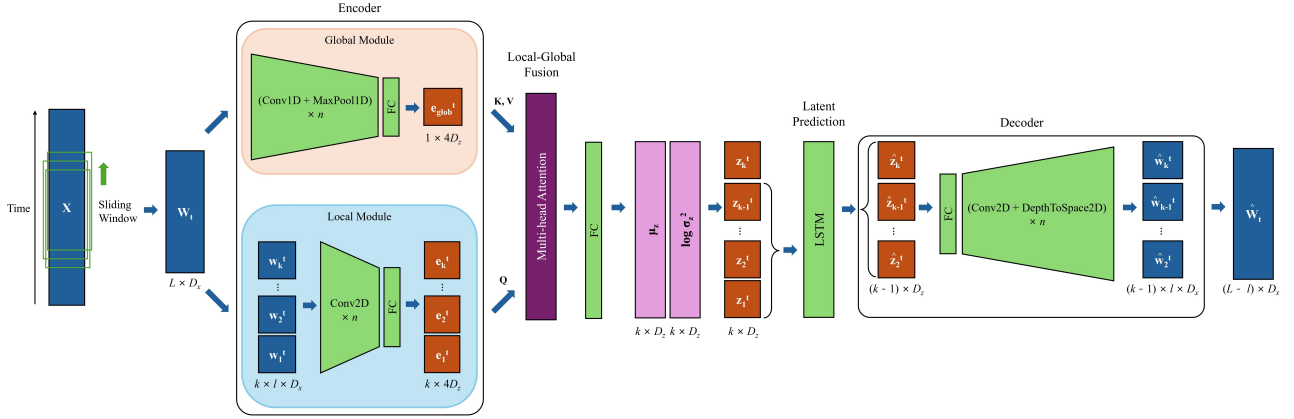


Figure 1. The architecture of our proposed model. The network processes the input time series using a dual-branch encoder (Global Module and Local Module). These features are fused by a Multi-head Attention layer to generate a comprehensive latent representation. LSTM layers model temporal dependency to predict the future latent sequence, which is then fed into the decoder for reconstruction.

V. The MA output, an enriched embedding sequence, is then processed to concurrently output the latent distribution parameters μ_t and $\log \sigma_t$ through a feed-forward network. Finally, the reparameterization trick yields the comprehensive latent representation sequence $\mathbf{Z}_t = [z_1^t, \dots, z_k^t]$.

3.3. Latent Prediction with LSTM

The latent sequence $\mathbf{Z}_t$ is fed into an LSTM network, which is crucial for modeling the temporal dependencies within the latent space and performing sequence prediction. The LSTM is trained in a sequence-to-sequence manner, taking the preceding $k-1$ elements of the latent sequence, $[z_1^t, \dots, z_{k-1}^t]$, as input. Its primary function is to predict the subsequent $k-1$ elements of the latent sequence, denoted as the predicted latent sequence $[\hat{z}_2^t, \dots, \hat{z}_k^t]$. This prediction mechanism enforces the model to learn the temporal patterns of the data in the compressed latent space, making the prediction error highly sensitive to unexpected changes.

3.4. Decoder and Anomaly Detection

The decoder takes the predicted latent sequence $[\hat{z}_2^t, \dots, \hat{z}_k^t]$ outputted by the LSTM. The Decoder's role is to map these high-level latent predictions back into the original data space using a structure that employs 2D convolutional (2DConv) layers and 2D depth-to-space (DepthToSpace2D) blocks, thereby reconstructing the corresponding portion of the input window, $\hat{\mathbf{W}}_t = [\hat{w}_2^t, \dots, \hat{w}_k^t]$.

During inference, the anomaly score is calculated by the sum of squared error between the predicted window $\hat{\mathbf{W}}_t$ and the ground truth window $[\mathbf{w}_2^t, \dots, \mathbf{w}_k^t]$, defined by the

following equation.

$$d_t = \sum_{i=2}^k \|w_i^t - \hat{w}_i^t\|_2^2. \quad (4)$$

If the computed anomaly score exceeds a predetermined threshold τ , the corresponding time interval is flagged as an anomaly.

For practical relevance in real applications where the detection of entire anomaly segments is prioritized rather than individual points, we adopt a modified segment-based evaluation strategy (Xu et al., 2018). This approach adjusts the point-wise metrics: if the detector triggers an anomaly signal for any point within a contiguous anomaly segment defined by the ground truth, the entire segment is classified as correctly detected. Points outside anomaly segments are evaluated normally. Evaluation metrics are then calculated based on these adjusted signals.

3.5. Loss Function

The model is trained by minimizing the overall loss function, $\mathcal{L}_{\text{total}}$, which is derived from the negative ELBO of the β -VAE. The loss function $\mathcal{L}_{\text{total}}$ is a weighted sum of the reconstruction loss $\mathcal{L}_{\text{recon}}$ and the KL divergence loss $\mathcal{L}_{\text{KL}}$.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{recon}} + \beta \mathcal{L}_{\text{KL}}. \quad (5)$$

By assuming a Gaussian distribution for the decoder output, the reconstruction loss takes the form of a scaled Mean Squared Error (MSE). A key modification is that the loss calculation is specifically tailored to the LSTM prediction mechanism. Since the LSTM only predicts the subsequent $k-1$ latent segments $[\mathbf{z}_2, \dots, \mathbf{z}_k]$, the $\mathcal{L}_{\text{recon}}$ is calculated

exclusively for the corresponding predicted segments of the window. Let $\mathbf{w}_i$ be the i -th ground truth segment and $\hat{\mathbf{w}}_i$ be the i -th predicted segment. For a batch size B , $\mathcal{L}_{\text{recon}}$ is calculated only for segments $i = 2, \dots, k$.

$$\mathcal{L}_{\text{recon}} = \frac{1}{B(k-1)} \sum_{b=1}^B \sum_{i=2}^k \|\mathbf{w}_{b,i} - \hat{\mathbf{w}}_{b,i}\|_2^2. \quad (6)$$

4. Data Preparation

4.1. Data Description

We utilize the WESAD (Wearable Stress and Affect Detection) dataset (Schmidt et al., 2018), a multimodal benchmark comprising physiological signals recorded from 15 participants during three affective phases: ‘baseline’, ‘stress’, and ‘amusement’. Each participant was monitored using two synchronized wearable devices: the RespiBAN Professional chest sensor (ECG, EDA, EMG, RESP, TEMP, ACC at 700 Hz) and the Empatica E4 wristband (BVP at 64 Hz, EDA at 4 Hz, TEMP at 4 Hz, ACC at 32 Hz). The physiological features relevant to stress detection are detailed in Table 1.

Table 1. Physiological signal features of WESAD dataset.

Feature	Description
ECG	Measures high-fidelity cardiac electrical activity, used to derive Heart Rate Variability (HRV).
BVP	Measures peripheral blood flow changes, used as an indicator of Heart Rate (HR).
EDA	Measures Electrodermal Activity (skin conductance) and Galvanic Skin Response (GSR), indicating sympathetic arousal.
EMG	Measures muscle electrical activity, indicating muscle tension.
RESP	Measures breathing rate and patterns.
TEMP	Measures skin temperature changes, regulated by the autonomic nervous system.
ACC	Captures motion and activity across three axes, used for motion artifact correction.

4.2. Data Preprocessing

The raw WESAD sensor data, characterized by heterogeneous sampling rates and inherent noise, is processed using the NeuroKit2 library to create a unified, multivariate time series (Garg et al., 2021). This involves resampling all signals to a consistent sampling rate of 4Hz and deriving 67 time-domain features, which are used instead of raw values. For ground truth mapping, ‘baseline’ and ‘amusement’ states are aggregated as the non-stress class, while the ‘stress’ state serves as the target anomaly class.

The data is partitioned using a subject-level split, dedicating data from 10 participants for training, and reserving data from the remaining 5 participants for the test set. The

training set is constructed solely from the non-stress state segments to establish the normal manifold. To ensure numerical stability, min-max scaling is applied, with the minimum and maximum values computed only from the training set and subsequently applied across the entire dataset.

To prepare the data for the proposed model architecture, the continuous, multivariate time series is segmented into overlapping windows $\mathbf{W}_t$ of length L , creating the final input format. Furthermore, a randomly drawn 10% of the total generated windows within the training set is reserved to form the validation set, which is excluded from the corresponding training phase.

5. Experiments and Results

5.1. Experimental Setup

The model was implemented using the PyTorch framework. For optimization, the Adam optimizer was utilized with a learning rate of 4×10^{-4} and moment decay rates set to (0.9, 0.95). Training was guided by early stopping (patience = 10) and a cyclical KL annealing schedule (Fu et al., 2019). This schedule began with a grace period of 25 epochs, during which the KL weight β was linearly ramped from 0 to 1×10^{-8} . Following this period, β was ramped cyclically over 25-epoch periods, increasing linearly from 1×10^{-8} to 1×10^{-2} . All models were trained for a maximum of 100 epochs.

The study employed several architectures for comparative analysis. The baseline models, adopted from existing literature, include the MA-VAE and the VAE-LSTM architecture utilizing only the Local Module for encoding. To demonstrate the impact of the Global Module and the Multi-head Attention (MA) fusion, we performed an ablation study comparing the VAE-LSTM architecture where local and global features are combined via linear concatenation against our proposed architecture where local and global features are combined via Multi-head Attention.

The window parameters were fixed across all architectures. For VAE-LSTM models, the window length L was set to 576, the segment length l was set to 48, and the window shift was fixed at 1. The baseline MA-VAE model was tested using a window length of 576 and a window shift of 48. The final anomaly scores were evaluated on the held-out test set using standard anomaly detection metrics: accuracy, precision, recall, and F1 score. All metrics are evaluated at the threshold that gives the best F1 score.

5.2. Results and Comparative Analysis

Table 2 summarizes the comparative performance results for the models and the number of model parameters. This analysis systematically validates the efficiency and contribution

Table 2. Performance comparison of models under fixed window parameters, including model size.

Method	# Parameters	Accuracy	Precision	Recall	F1
MA-VAE	10.1M	0.641	0.506	0.988	0.669
VAE-LSTM, Local Module Only	1.66M	0.912	0.953	0.795	0.867
VAE-LSTM, Local + Global (Linear Concat)	2.25M	0.890	0.783	0.964	0.864
VAE-LSTM, Local + Global (MA)	2.26M	0.956	0.912	0.971	0.941

of the LSTM prediction mechanism, the Global Module, and the proposed Multi-head Attention (MA) fusion strategy.

The results confirm the necessity of temporal prediction over reconstruction-only methods. The baseline MA-VAE achieved the lowest F1-score (0.669). Despite registering the highest recall (0.988), its poor precision (0.506) validates the model’s inability to robustly discriminate anomalies from noise. Crucially, the MA-VAE model is the largest configuration, utilizing 10.1M parameters, yet yields the lowest performance, highlighting poor efficiency.

The introduction of the LSTM prediction mechanism, using the VAE-LSTM architecture utilizing only the Local Module, immediately raised the F1-score to 0.867. This improvement is primarily attributed to achieving the highest precision (0.953) among all configurations, confirming that modeling temporal dynamics in the latent space is critical for robust anomaly discrimination. Furthermore, this VAE-LSTM local module achieves this superior discrimination with a significantly smaller memory footprint, requiring only 1.66M parameters.

The ablation study validates the efficacy of the proposed fusion mechanism. Combining Local and Global features via Linear Concatenation resulted in performance stagnation (F1 0.864). While it yielded high recall (0.964), the degraded precision (0.783) suggests that simple concatenation introduces contextual interference, hindering the prediction task.

In contrast, our proposed VAE-LSTM architecture where local and global features are combined via Multi-head Attention achieved the highest performance across all metrics, achieving the best F1-score (0.941) and accuracy (0.956). This represents a substantial improvement, demonstrating the model’s ability to maintain high Recall (0.971) while sustaining strong Precision (0.912). Remarkably, this superior performance was attained with a negligible increase in model complexity, utilizing 2.26M parameters, confirming the high efficiency of the MA-based dynamic fusion.

The findings from our comprehensive ablation study confirm that the dynamic and weighted fusion of multi-scale features via the Multi-head Attention mechanism is essential for achieving the optimal balance between correctly identifying anomalies and minimizing false positives. The low parameter overhead associated with the MA fusion

mechanism underscores the high efficiency of our proposed architecture. The superior performance of our proposed architecture validates the benefit of integrating long-term context into the prediction of fine-grained local temporal patterns for robust time series anomaly detection.

6. Conclusion and Future Work

6.1. Conclusion

Our results highlight the clear limitation of reconstruction-based architectures such as MA-VAE, whose poor precision confirms the inadequacy of simple reconstruction methods for robust anomaly discrimination. In contrast, our VAE-LSTM model with its dual-branch encoder and Multi-head Attention (MA) fusion achieves stronger and more balanced anomaly detection performance (F1 0.941). The proposed model successfully integrates long-range contextual patterns with fine-grained temporal cues, overcoming the challenges identified in baseline methods. Importantly, this superior performance is achieved while remaining highly compact in parameter count (2.26M), validating the efficiency of our dynamic fusion approach. Characterized by its efficiency and balanced performance, our proposed VAE-LSTM architecture is suited as a robust and practical solution for real-time physiological stress monitoring.

6.2. Future Work

While this work evaluates the model under a fixed window configuration, the architecture is inherently compatible with a wide range of temporal resolutions and shifts. A natural extension is to systematically vary window length and window shift to examine how the F1-score and related metrics change across different temporal granularities. This would allow us to assess the robustness of the proposed model and clarify the practical trade-offs between temporal resolution and computational efficiency for real-world use.

Moreover, since the model already delivers strong performance with a compact parameter budget and fast training, another direction is to scale the architecture in a controlled manner, such as deepening the local and global branches, expanding the latent space, or adopting more expressive attention mechanisms tailored to physiological time-series patterns, while still maintaining real-time feasibility.

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